

# RegRed a new global database of regional red list assessments

Florencia Grattarola, Gabriel Ortega, Ivo Kadlec, Daniela Mellado, Safya Praleskouskaya, Carmen Soria, Kateřina Tschernosterová, Adam Uličný, Kristýna Vlasáková, Petr Keil

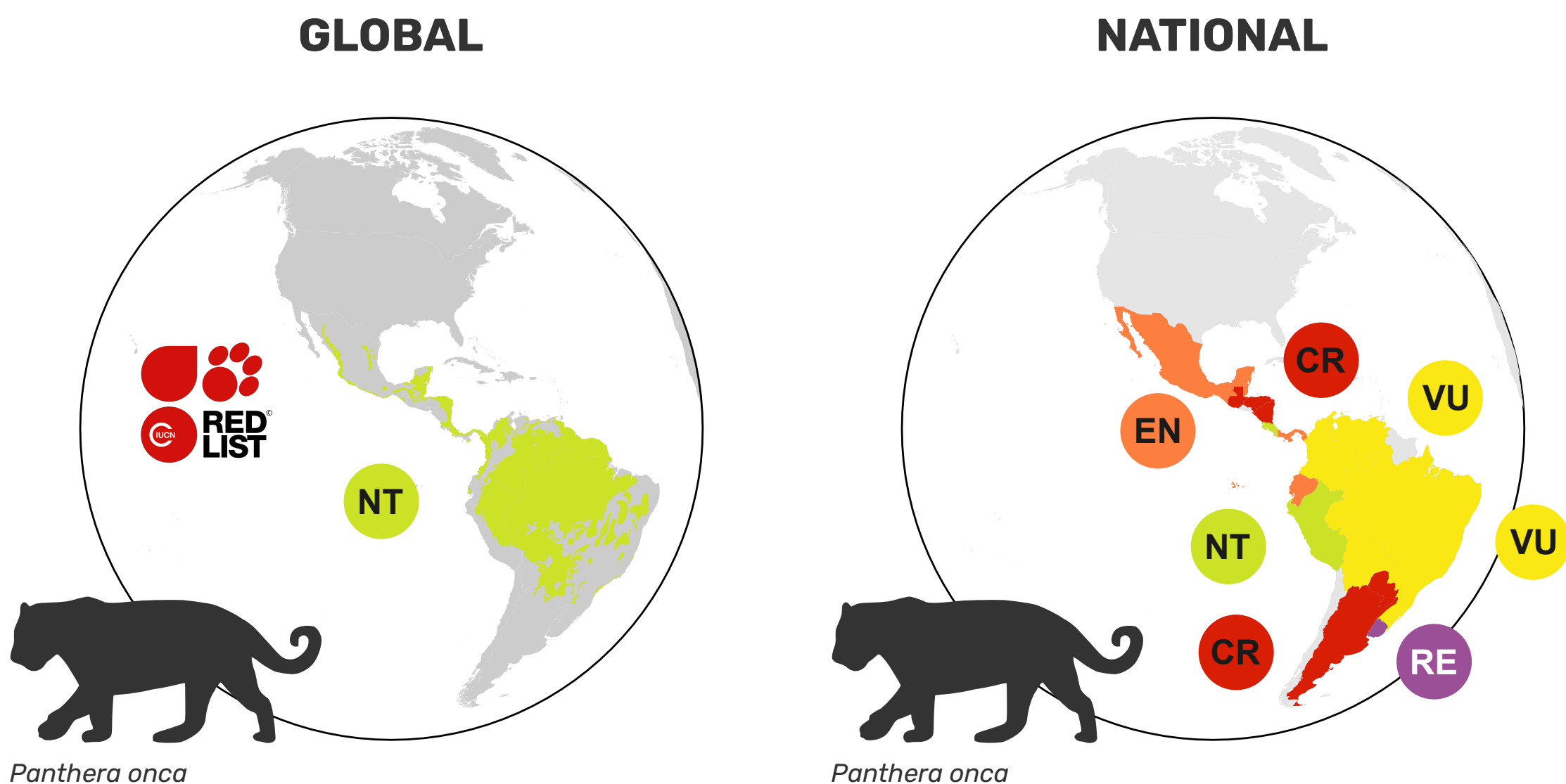
Modelling of Biodiversity (MOBI) Lab, Faculty of Environmental Sciences, Czech University of Life Sciences Prague

## Why create RegRed?

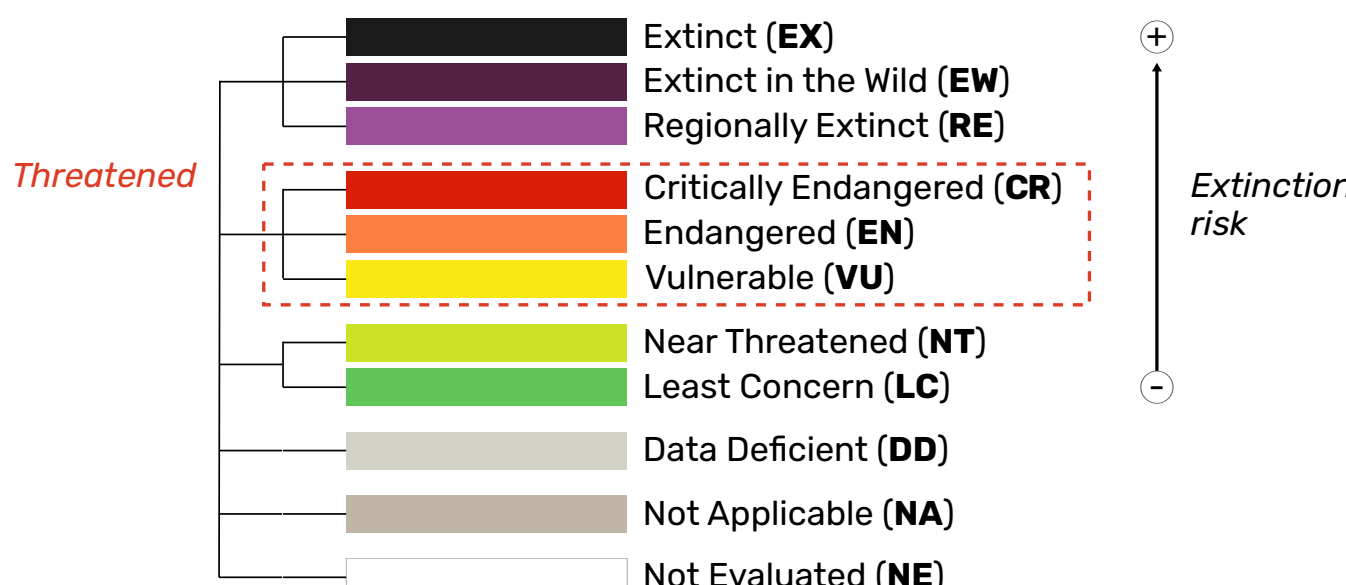
Global evaluations, like the **IUCN Red List of Threatened Species™**, are available across a species' entire range. However, while a species may be globally safe, it can face local extinction or significant population decline within specific countries.

Luckily, many countries and regions also produce their own threat assessments, often compiled into regional or national red lists.

**RegRed** is a project dedicated to developing a comprehensive, accessible database of regional red lists, to document species extinction risk status at the regional, national, and subnational levels. In this first phase, we are focused on tetrapods.



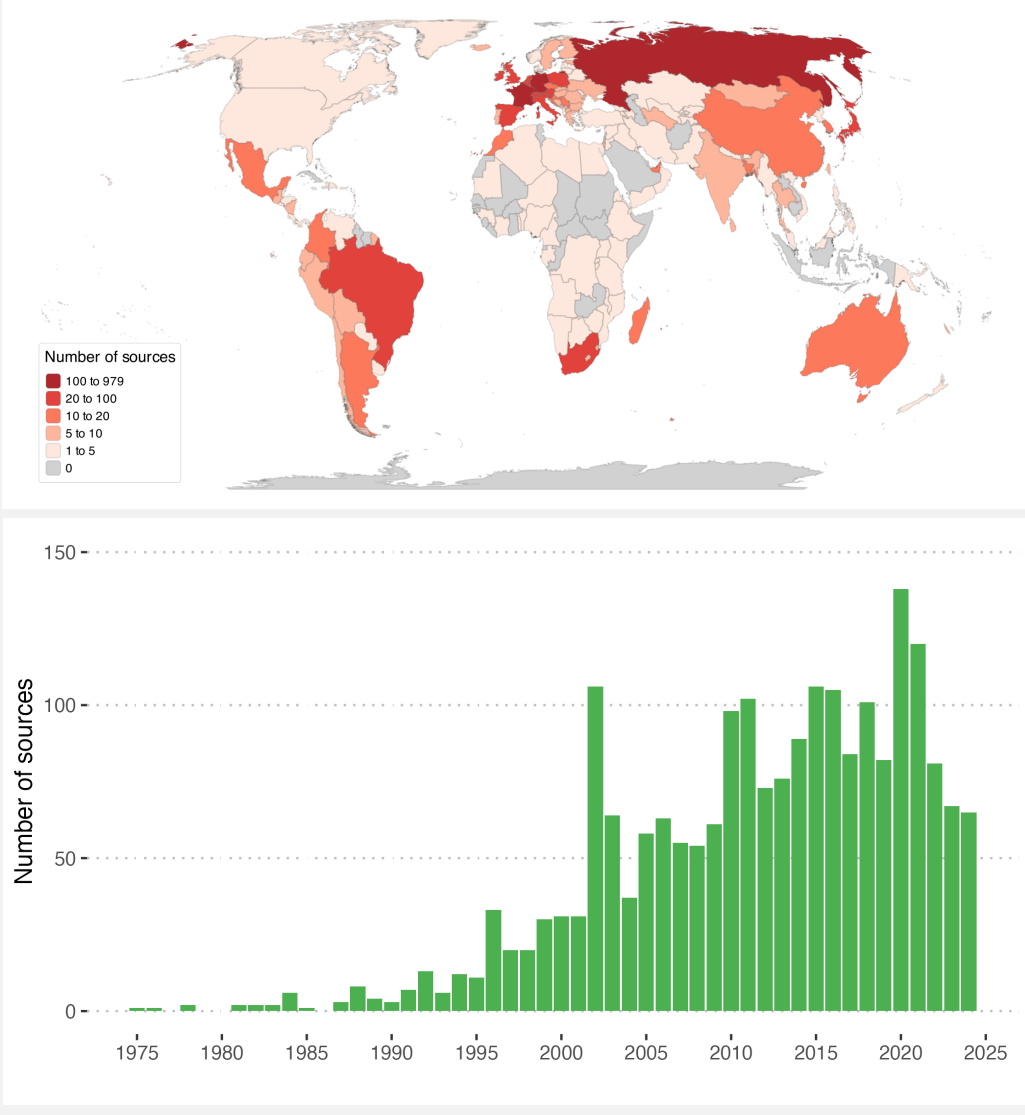
Guidelines for application of IUCN Red List criteria at regional and national levels: version 4.0



## How did we create RegRed?

### 1 DATA SOURCES

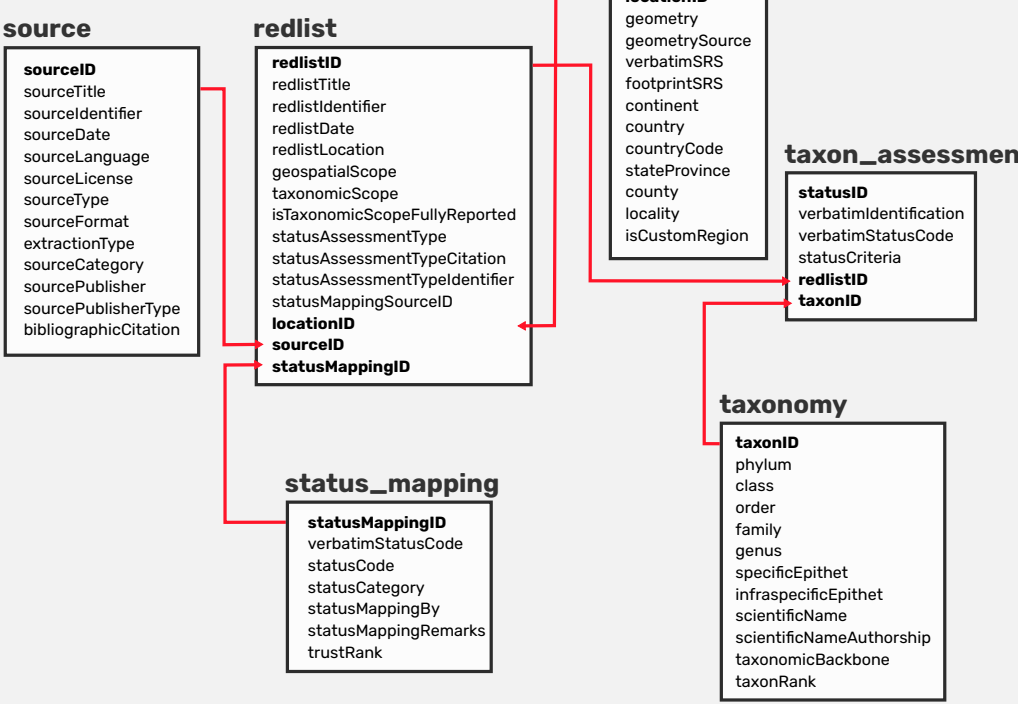
We searched and compiled a list of all relevant sources for all possible taxa around the world and published them in a metadata data paper.



### 2 GEOSPATIAL DATABASE

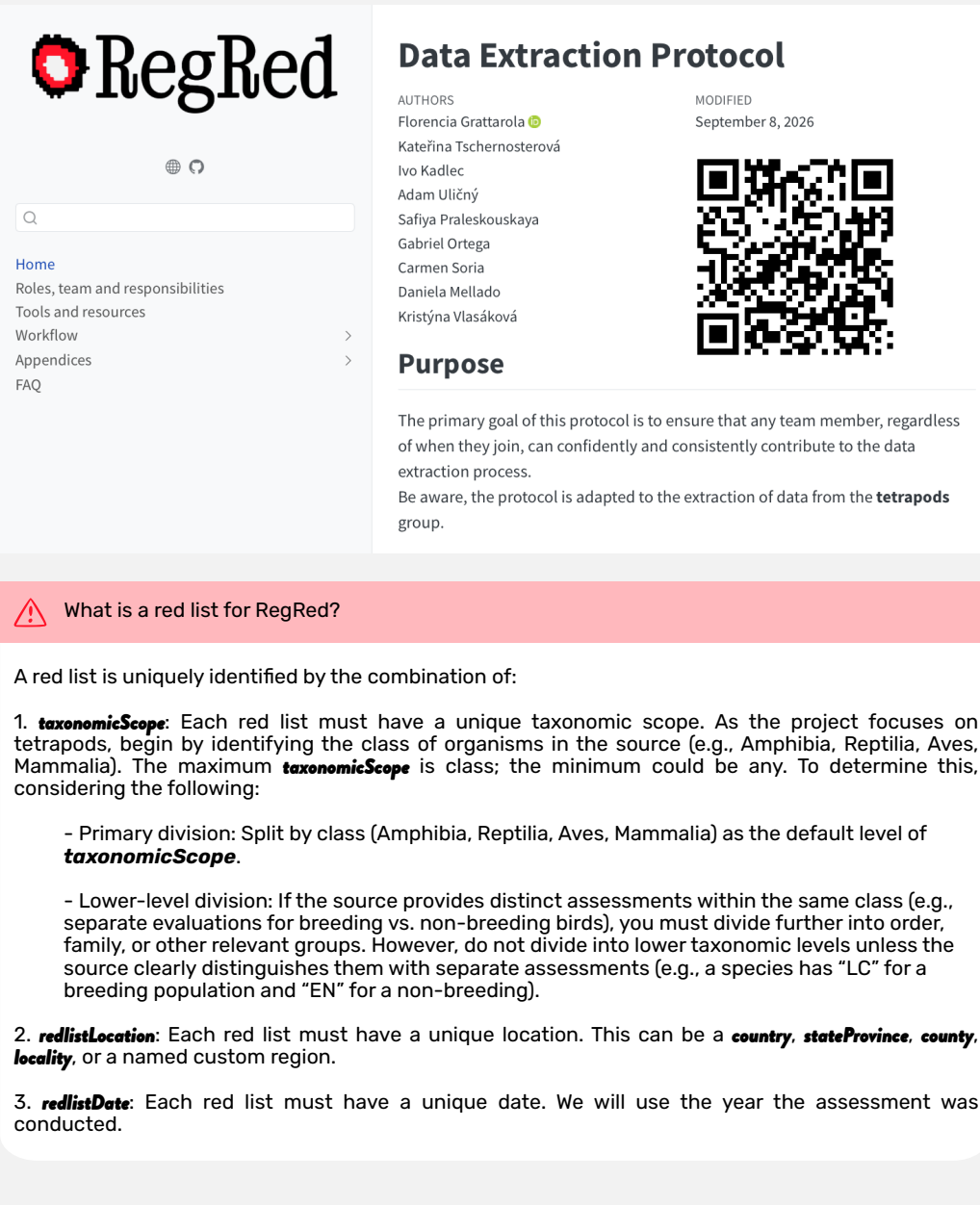
We designed a geospatial relational database (PostgreSQL) with five main tables that follows biodiversity standards and is highly flexible to different taxonomies, locations, and threat statuses.

Tables (simplified)



### 3 DATA EXTRACTION PROTOCOL

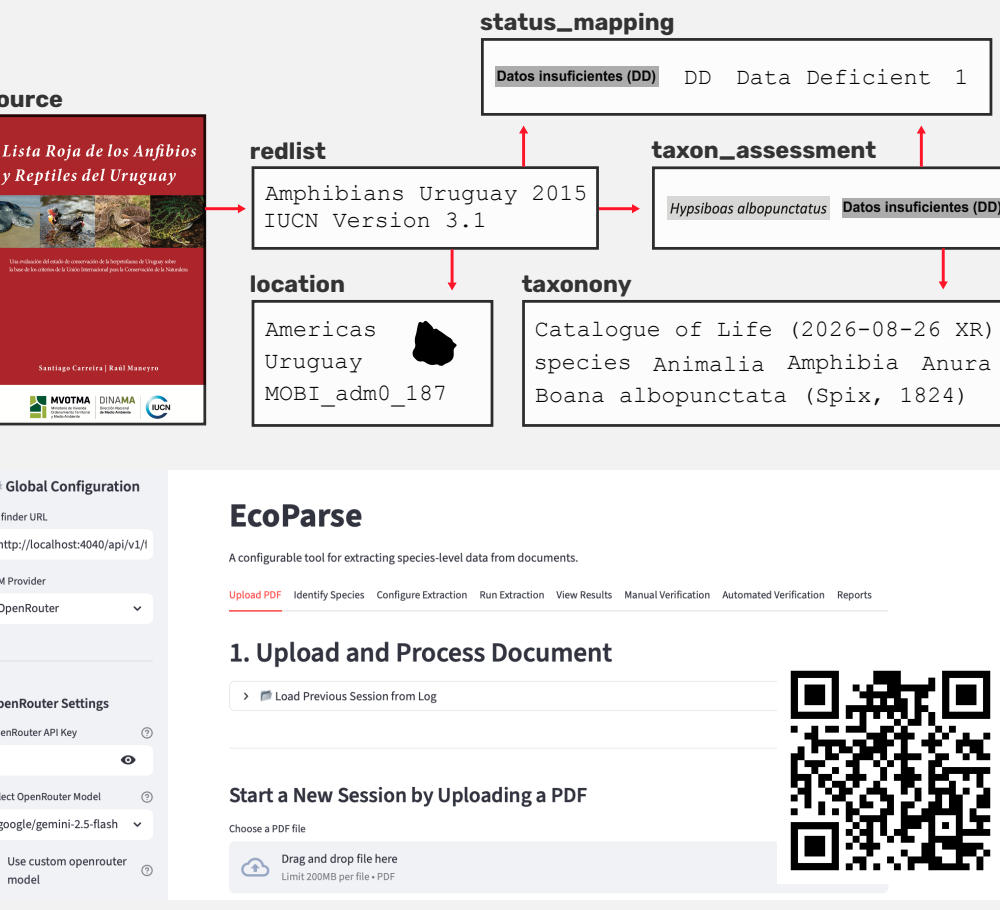
We created a protocol to extract data from each source using spreadsheets.



### 4 DATA EXTRACTION

We extracted data from tetrapod sources manually and also in an automated way using *EcoParse*, a flexible tool designed to extract species-level data from PDFs.

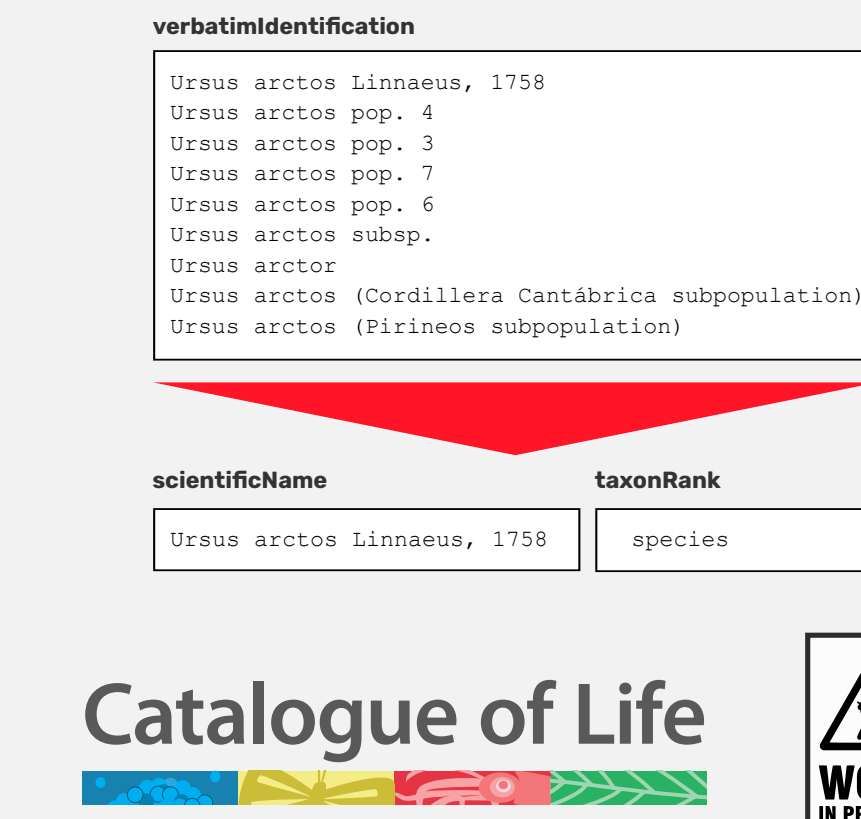
Extraction example (simplified)



### 5 TAXONOMIC HARMONISATION

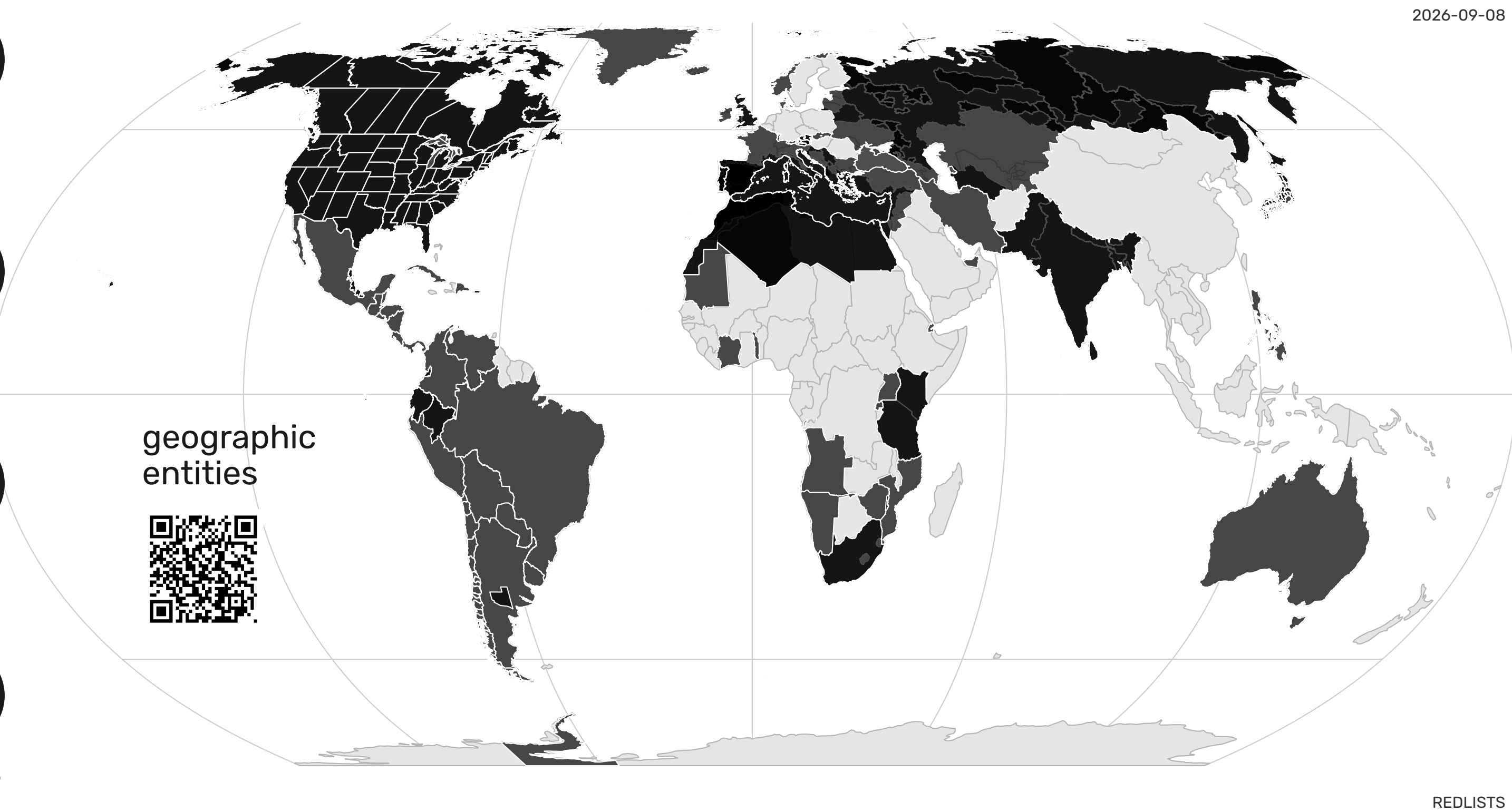
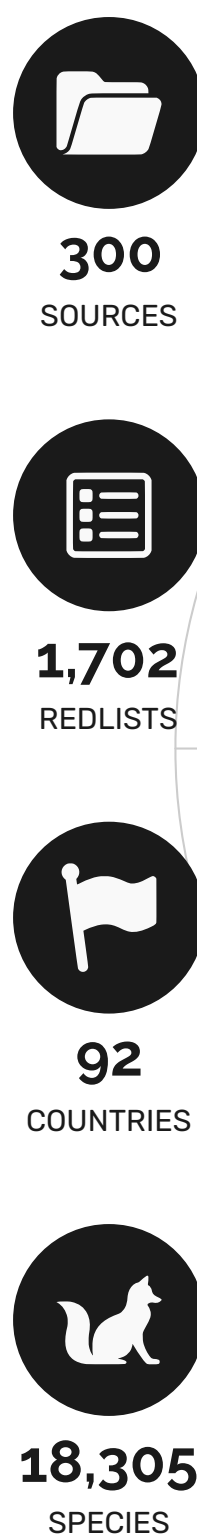
We will harmonise species names using the Catalogue of Life, the most complete authoritative list of the world's species.

Harmonisation example (simplified)



## What have we accomplished so far?

We have processed **59.1%** of the sources so far, accounting for **1,702** red lists and covering **92** countries, **19** of which also have red lists at the sub-national level. From an initial list of **32,423** species names, we have preliminarily matched these to **18,305** unique species (with **6,847** unresolved names). We have standardised **114,906** threat status assessments.



## What's next?

- Process the remaining 40.9% of sources.
- Finalise the geographic entities (including custom regions).
- Harmonise all species names.
- Develop the front end of the RegRed website.
- Publish a data paper to release the first version of the database.
- Serve the first version of the database online.

The first version will be released in 2027

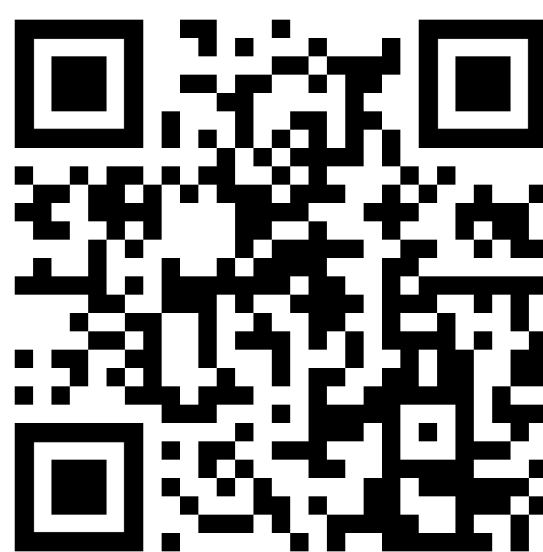


- Open access
- Geospatial database
- Standardised taxonomy
- Standardised threat statues

## Stay in touch



Metadata paper



GitHub project



regred@fzp.czu.cz

**Credits** Icons from Freepik and Magnific via Flaticon.com. *Panthera onca* by Gabriela Palomo-Muñoz (CC-BY-NC) via Phylopic. Map by Gabriel Ortega. All other figures were created by Florencia Grattarola. RegRed logo design Eugen Korda.

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**RegRed's main contact** Florencia Grattarola [grattarola@fzp.czu.cz](mailto:grattarola@fzp.czu.cz) [flograttarola.com](https://flograttarola.com)



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CTU Faculty of Environmental Sciences

